DISASTERS DOOR: INSPIRING YOUTH TO BUILD RESILIENCE THROUGH GAMIFIED PREPAREDNESS

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

Over the past years, there has been a growing concern of disaster preparednessamong countries that are always being hit by occurrences of natural hazards particularly in the Philippines. The country itself, geographically situated, becomes highly favorable to the conducive environment of typhoons, earthquakes, floods, and volcanic eruptions that consequently cause vast short-term devastation and long-term economic challenges to the communities (CFE-DM, 2020). Therefore, schools have been equipped with a disaster risk reduction and management (DRRM) education to enable the youth to learn how to survive and be aware of their disaster risk. The current changes in education are also seen to be caused by the growing use of technology-based strategies. Gamification in the classroom is one of the aspects that have recently experienced the influx; it has been demonstrated to drive higher engagement, knowledge retention, and motivation as compared to the traditional lecture forms (Subhash and Cudney, 2018; Koivisto and Hamari, 2019). The very fact that it is done demonstrates that educators seek different ways of preparing students better against life and academic challenges, such as disaster preparedness.

Even with the school-based DRRM programs, one major problem still exists; students tend to forget or not use the lessons on disaster preparedness in the real case of an emergency. The present activities, including lectures and drills on earthquakes,

are typically presented based on passive and repetitive approaches that do not keep students interested or ensure their long-term memory (Ahn, Johnsen, and Ball, 2021). It has been discovered that the learning process can become passive, resulting in a shallow knowledge that cannot be applied in the crisis situation (Culbertson et al., 2019). This issue is especially worrisome concerning the Philippines where young learners constitute one of the most vulnerable populations in times of calamities because of their age and their reliance on adults. Devoid of a more living and interactive methodology, one may end up producing learners who are familiar with preparedness procedures in theory but have no confidence in their ability to put them into practice.

To address these issues, the proposed study presents Disaster Doors, the online educational game that strengthens the understanding of disaster preparedness among junior high school students by moving away from inactive and drill-based learning to interactive and games-based learning. The game provides two modes, including fantasy-style adventure which combines elemental trial doors to earthquakes, floods, and fire challenge, and real-life simulation whereby the student players are taken through disaster challenges as high school learners. Helping students rehearse disaster response challenges like performing a duck, cover, and hold, turning off any electricity in the event of flooding, or putting out fires, the game offers an experiential learning environment that facilitates increased understanding and retention (Ahn et al., 2021). By means of this, the research will show how preparing students to disaster can be more interesting, topical, and memorable through gamification methods.

The contribution that this study makes is in filling the gap between the traditional disaster preparedness education and the novel and innovative technologies-driven approaches to learning. With the junction on the junior high school students, the study has indicated the developmental phase wherein the students start developing a

sense of responsibility and being independent, so preparedness lessons become more effective. The project gives the students the strength to be active participants in their safety as opposed to passive receivers of information. On a bigger level, the research is also engaged in the advancement of a culture of safety and resilience within schools and communities, which is also part of the global frameworks like the Sendai Framework of Disaster Risk Reduction (UNDRR, 2021). Moreover, it contributes to the emerging literature on gamification in education, demonstrating how the digital learning tools can be successfully used as supplement to the traditional teaching and as the means to resolve the burning issues of the real world (Alsawaier, 2019; Xu et al., 2020). Finally, this paper highlights the concept that disaster preparedness cannot merely be taught as a compliance practice but should be made an attractive, valuable experience that teaches the student the skills and the confidence to act constructively in the event of an emergency situation.

Statement of the Problem

The researchers are conducting this study with the goal of developing Disaster Doors. This system is designed to make disaster preparedness education more interactive and engaging compared to the usual drills and lectures. The game will help students practice what to do during emergencies such as earthquakes, floods, and fires in a simulated environment.

Specially, the study sought to answer the following questions:

1. How effective is the Disaster Doors system in terms of:

    1.1 Game design;

1.2 Game mechanics?

2. To what extent does the system improve disaster preparedness among learners in terms of:

2.1 Knowledge acquisition;

2.2 Skill application; and

2.3 Decision-making during simulated scenarios?

3. How to evaluate the level of acceptability of the Disaster Doors system based on ISO/IEC 25010 software quality standards in terms of:

3.1 Functional Suitability;

3.2 Performance Efficiency;

3.3 Compatibilty;

3.4 Usability;

3.5 Reliability;

3.6 Security;

3.7 Maintainability; and

3.8 Portability.

Objectives of the Study

The general objective of this study is to develop Disaster Doors, an interactive game that will help junior high school students learn disaster preparedness in a fun and engaging way through gamified simulations.

Specifically, this study sought to:

1. To improve students knowledge, skills ,and decision-making in responding to disasters.

2. To make disaster preparedness education more engaging through gamified simulations.

3. To evaluate the system's effectiveness and quality based on ISO/IEC 25010 of software quality standards, , which determine the degree of satisfaction of the users based on the implied needs of the users, such as the following:

- 3.1 Functional Suitability.
- 3.2 Performance Efficiency.
- 3.3 Compatibility.
- 3.4 Usability.
- 3.5 Reliability.
- 3.6 Security.
- 3.7 Maintainabilty.
- 3.8 Portability.

Significance of the Study

This study explores the effectiveness of Disaster Doors, an interactive 3D simulator game, in enhancing disaster preparedness among secondary school learners. By immersing learners in realistic emergency scenarios common in the Philippines,

such as: earthquakes; fires; and floods. This study evaluates how gameplay can improve awareness, response skills, and overall preparedness in real-life disaster situations.

The findings of this study will be valuable to:

To the Educators. This study provides insights on integrating Disaster Doors into disaster risk reduction curriculum, enhancing teaching methods and student engagement. This is useful in fulfilling the objectives of the educators to teach learners faster in a more fun and interactive way, especially for Science educators based on their curriculum.

To the Learners. Learners will gain practical knowledge and skills to respond in emergencies through engaging gameplay, ultimately contributing to their safety and well-being. Furthermore, this is a fun way for them to understand disaster risks and prevention without neglecting its purpose.

To the Disaster Agencies. The game is used as a tool to promote disaster awareness and preparedness among communities, supporting agency initiatives.

To the Game Developers. Insights from this study can inform the design of more effective educational games, enhancing the learning experience for users with the focus on helping the bigger demographic scale, and on what is in demand.

To the Parents and Guardians. By educating children on disaster preparedness using this more advance way, parents and guardians can feel more confident in their children's ability to respond to emergencies without too much pressure on their end.

To the Community Leaders. The study's findings can inform community-based disaster risk reduction initiatives, enhancing overall community resilience.

To the Future Researchers. This study serves as a foundation for further research on the effectiveness of interactive games in disaster education, providing a framework for future studies.

Scope and Delimitation

The scope and delimitation of this study are presented to clarify its boundaries and focus. This section outlines the coverage of the research, including the participants, the specific areas of disaster preparedness being assessed, and the limitations set to maintain the study's feasibility and direction.

Login/Sign Up. Enable players to create an account, which is required for auto saving the game progress and loading. Log in function for accessing or entering the game. Sign up to create a new account. The registration of user authentication for database handling using Firebase.

Game menu. Consists of three buttons: Start, Options, and Exit. The start button is required for entering the game. The options consist of adjustable customization based on users' preferences. And, exit for quitting the game.

Disaster Events. The game highlights disaster events such as earthquakes, floods, and house fires that trigger upon entering the trial doors. The events test players' knowledge in preparing for safety during disasters.

Progressive Difficulty System. The game adjusts the complexity of disaster scenarios as the player advances through stages. Event triggers automatically update the game state to the next predefined difficulty and are controlled by a difficulty

manager. Moreover, it is essential to ensure that players gradually develop disaster preparedness skills through high-stress situations.

Interactive Elements. The game also includes fun facts and quizzes to reinforce disaster preparedness and knowledge of different disaster events while keeping the experience enjoyable. The quiz is inside a quiz module of the program. The fun facts are an info pop-up component triggered by events.

Time Pressure System. Composed of a Countdown Mechanic and a Panic Gauge. The countdown administers quick decision-making, while the panic gauge portrays player stress, increasing difficulty and engagement during disaster scenarios.

Sequential Narrative System. Delivers the storyline through sequential illustrated panels. Each slide is displayed through the UI Panel System, while event triggers control when fun facts, quizzes, or dialogues appear. It handles transitions between slides using the Scene Manager and State Machine.

While the Disaster Doors aims to enhance disaster preparedness and knowledge acquisition of disasters, the study also acknowledges the following limitations:

1. The game's system difficulty is preset, and players are restricted from configuring it.
2. The event handler is limited to three disaster events: Earthquake, Flood, and Fire. No extensible event handler or modular plug-in system exists for adding new disaster types without modifying core code.
3. The system is only compatible with personal computers, laptops and mobile platforms. No abstraction layer is implemented for Console Application Interfaces or Virtual Reality frameworks.

4. Multiplayer mode is not supported. This is a single-player game, purposefully designed as an individual player instance for a better experience. No multiplayer synchronization or client-server architecture is implemented.
5. The simulation is simplified to basic script interactions and does not cover advanced emergency preparedness procedures. It only provides basic survival actions satisfactory for knowledge acquisition and disaster experience.
6. It is a hardware resource dependence. The system does not implement dynamic asset scaling or performance optimization layers. On low-end devices, memory leaks, frame drops, or reduced rendering quality may occur due to hardware limitations.
7. The panic tracking system is simplified. The panic gauge is implemented as a linear variable that only increments on events. No advanced programming engine is supported, such as Artificial Intelligence(AI) or psychological profiling.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of relevant literature, and related studies were reviewed to gain a clearer perspective and direction for the study, as well as to establish an adequate background for the study. It includes the research problem, methodology,

key findings and conclusions from previous research, theoretical expectations regarding the development of the current study, and information gathered from various scholarly sources. It also includes the conceptual framework and the research paradigm of the study.

Relevant Theories

Gamification in Education. According to Koivisto and Hamari (2019), gamification is the use of game design elements in non-game contexts to increase motivation, engagement, and learning. In educational settings, gamification enhances active participation by embedding mechanics such as points, feedback, and challenges. Their study emphasizes that gamification not only improves cognitive engagement but also strengthens emotional involvement and long-term retention. Within the context of Disaster Doors, gamification ensures that disaster preparedness is both interactive and enjoyable, fostering students' motivation to practice and apply survival skills.

Experiential Learning. According to Kolb and Kolb (2021), experiential learning is a process whereby knowledge is created through the transformation of experience. It follows a four-stage cycle: concrete experienc e, reflective observation, abstract conceptualization, and active experimentation. Recent studies highlight that experiential learning is most effective when learners face real-world problems requiring immediate responses. In Disaster Doors, this is achieved by simulating earthquakes, floods, and fires, giving students opportunities to practice decisions, reflect on consequences, and transfer skills to real-life disaster contexts.

Self-Determination Theory. According to Ryan and Deci (2020), Self-Determination Theory (SDT) explains that intrinsic motivation grows when three basic psychological needs—autonomy, competence, and relatedness—are met. In learning environments, SDT emphasizes that students thrive when they are free to make choices, feel capable in completing tasks, and connect their efforts to a larger social pur pose. Disaster Doors incorporates SDT by letting learners make independent choices during gameplay, gain mastery through disaster-response challenges, and recognize their role in promoting community resilience.

Educational Escape Rooms. According to Fotaris and Mastoras (2019), educational escape rooms provide immersive, problem-solving experiences where learners collaborate under time constraints to achieve specific learning outcomes. Their research highlights that the narrative-driven and puzzle-based nature of escape rooms supports critical thinking, teamwork, and knowledge application. In Disaster Doors, this approach is applied through adventure-based missions and realistic disaster simulations.

Related Literature

Gamification has become a powerful tool in modern education, offering a way to transform traditional learning into an interactive and engaging experience. By incorporating elements such as points, levels, feedback, and storytelling, gamification encourages active participation and deepens student engagement.

Foreign Literature

According to Kim, S., & Park, J. (2021), a VR-based earthquake preparedness program in South Korea provided immersive training to middle school students. Learners experienced shaking simulations, evacuation procedures, and hazard identification tasks in a controlled VR classroom environment. Pre- and post-tests revealed a substantial increase in procedural memory, indicating VR's strong potential for reinforcing real-world readiness.

Nakamura, Y., & Tanaka, K. (2020) developed "SafeCity," a serious game for junior high school students that addressed earthquakes, floods, and fires. Using city-mapping and decision-tree gameplay, students practiced risk assessment and emergency prioritization. The game's evaluation showed higher disaster literacy scores and improved collaborative problem-solving skills, emphasizing the value of scenario-based multi-hazard learning.

Rahman, F., & Utami, R. (2022) introduced a point-based gamified system during tsunami evacuation drills in Aceh. Students earned points for identifying correct routes, assisting peers, and completing safety checklists. Compared to traditional drills, participants reported higher motivation levels and demonstrated quicker response times, proving gamification's role in strengthening behavioral readiness.

Miller, T., & Johnson, P. (2019) implemented an AR fire evacuation application in Australian high schools, overlaying virtual fire and smoke effects onto the physical classroom. Students navigated safe exit strategies while responding to changing hazards. Assessment results indicated improved hazard detection skills and reduced hesitation in simulated evacuations.

Smith, L., & Carter, B. (2023) created “StormShield,” an educational tablet game teaching middle school students in Florida about hurricane tracking, home preparation, and evacuation planning. Players managed resources and made real-time decisions as storms approached. The study’s findings showed a significant boost in understanding meteorological warnings and a greater likelihood of following preparedness guidelines in future events.

Local Literature

According to Dela Cruz, M. & Santos, R. (2021), gamified learning has emerged as a transformative approach in Philippine disaster preparedness programs. Their study implemented a mobile game simulating typhoon and earthquake scenarios for senior high school students in Quezon City. Through challenges and rewards, learners practiced correct evacuation and first-aid procedures. Results showed that retention rates for correct safety protocols increased by 42% compared to traditional lectures. The researchers concluded that integrating gaming elements into DRR education fosters higher engagement and promotes more effective skill transfer in real-life emergencies.

Lacson, J.P., & Villaruel, C.A. (2020) developed “QuakeQuest PH,” a mobile application designed to teach earthquake safety through interactive scenarios and mini-games. Targeting public high school students in Manila, the app guided users through “duck, cover, and hold” drills, post-quake safety measures, and family reunification planning. Their evaluation revealed significant improvement in students’ confidence and readiness to act during seismic events, emphasizing the app’s role in bridging theoretical lessons and practical application.

Villanueva, M.C., & Bautista, K.L. (2022) created a 3D desktop-based simulation game replicating flood-prone neighborhoods in Bulacan. Players navigated

missions such as securing belongings, shutting down power, and leading family members to evacuation centers. The study, conducted among Grade 9 students, found that participants who trained with the simulation demonstrated faster and more accurate decision-making during a subsequent school flood drill compared to the control group.

Reyes, A.L., & Mendoza, E.C. (2019) designed a school-based escape room challenge centered on fire evacuation procedures for junior high school students in Cebu. Tasks involved identifying fire hazards, choosing safe exit routes, and operating fire extinguishers in a timed, gamified format. Results indicated increased hazard awareness and improved recall of fire safety steps one month after participation, supporting the integration of escape-room style learning in DRR programs.

Gonzales, P.R., & Lim, J.A. (2023) tested a gamified scoring and rewards system during community disaster drills in Pampanga. Households earned points for timely evacuation, proper packing of emergency kits, and safe transportation of vulnerable members. Survey data revealed increased participation rates and enthusiasm compared to previous years' drills. The study concluded that game elements in barangay-level activities encourage active citizen involvement and readiness.

Related Study

The related study introduces various methodologies that are helpful throughout the study. The author's name, participants, context, concept, and key findings of the included studies are summarized. The examined articles were published between the year 2019 to 2025.

Foreign Study

The study by Ryubikin et al. (2023) addressed the importance of earthquake education for foreigners intending to study or work in Japan, a country highly

vulnerable to natural disasters, particularly earthquakes and tsunamis. Given the unpredictability and severity of such events, survival largely depends on individuals' preparedness and response. The study emphasises that earthquake education enhances knowledge, readiness, and appropriate reactions before, during, and after an earthquake. Early exposure to this education enables foreign residents to develop risk awareness, improve survival skills, and build confidence in responding to seismic emergencies in Japan.

Saptaputra, S.K., et al. (2024). The study used a Participatory Rural Appraisal Approach, incorporating interactive tools like quizzes, flashcards, and simulations to educate participants on disaster mitigation, response, and evacuation. The study's pre- and post-tests showed a notable improvement in participants' disaster preparedness knowledge and skills. The findings also highlight the value of including hands-on disaster simulations in the school curriculum. As a result, the study offers a replicable model for disaster-prone areas, focusing on ongoing education and infrastructure development to foster community resilience.

Marahatta, D., et al. (2024). Game-based learning is an effective tool for enhancing disaster preparedness and response, particularly in resource-limited communities in the Global South. A study focused on educating children about landslides, earthquakes, and house fires demonstrated that this interactive approach significantly empowers marginalized school-aged populations who are often excluded from traditional disaster education. In Nepal—a country highly prone to natural hazards such as earthquakes, landslides, and fires—such educational methods are essential to equip communities with the knowledge and skills needed to reduce risks, respond effectively to emergencies, and build long-term resilience.

Garguilo, M.V. (2024). Serious games, such as educational escape rooms, offer an effective approach to engage youth in learning. These games prioritize educational outcomes over entertainment, promoting active participation and experiential learning. Students reported improved understanding of basic seismology concepts and increased awareness of disaster-related risks. Additionally, the activity fostered greater interest in science, particularly in geophysics.

Bai, S. (2024). Gamified approaches are categorised into seven types: tabletop games, serious games, scenario simulation games, virtual reality (VR) and mobile games, theme games, board games, and escape room games. Each method offers unique strengths: VR-based gamified learning provides immersive environments, serious games use storytelling to facilitate learning, and scenario simulations and escape rooms replicate real-life situations to develop practical skills. These diverse approaches not only make disaster education more engaging but also foster deeper learning and preparedness.

Local Study

A study conducted in selected communities in Central Luzon, Philippines examined residents' disaster knowledge and household preparedness. Results revealed that respondents possessed substantial disaster awareness and that households were generally ready to respond during emergencies. Statistical analysis indicated significant differences in disaster knowledge across communities, as well as a notable relationship between respondents' community and their level of disaster knowledge. Moreover, a moderate positive correlation was found between disaster knowledge and household preparedness. The study recommended strengthening community-based disaster education programs. (Dollete, 2020).

A study assessing disaster preparedness, disaster-related knowledge, and disaster adaptation among intermediate-level students found high levels in all three areas, indicating frequent demonstration of preparedness behaviors. Students valued sharing knowledge, learning from experts, and showed confidence in their school's readiness, as well as willingness to discuss disaster-related topics. Demographic factors showed no significant influence on preparedness and knowledge, though grade level significantly affected disaster adaptation. The findings highlight the importance of integrating inclusive and context-appropriate disaster risk reduction (DRR) programs into school curricula, while sustaining and strengthening existing preparedness initiatives. (Epal, 2024).

The study involving senior high school students of Eastern Samar State University assessed disaster-related knowledge, adaptation, preparedness and readiness, risk perception, and awareness. Despite being in a rural setting, students demonstrated readiness, awareness, and adaptability toward disaster risks. Significant differences were observed in disaster preparedness, readiness, adaptation, and knowledge based on sex, while grade level differences emerged only for disaster-related knowledge. All disaster risk reduction (DRR) categories varied significantly across academic strands, but family income showed no influence. The study recommended strengthening DRR education and initiatives in schools to further reduce students' vulnerability to disasters. (Lapada, 2022).

Studies on typhoon preparedness often indicate that individuals are inadequately prepared or demonstrate only low to moderate levels of readiness, despite the topic being part of the K–12 curriculum. To address this gap, a study developed a contextualized digital game for Grade 8 students as a supplementary learning tool to enhance decision-making and response knowledge during typhoons. The game was

found to be highly acceptable and effective in promoting disaster preparedness. Learners showed a marked improvement in their understanding and awareness after using the game, confirming its value as an educational resource that fosters active participation, motivation, and accurate disaster-related knowledge. (Tayone, 2024).

A study addressed the limited scope and frequency of school and office disaster drills, which are typically conducted only once a year and focus mainly on earthquakes and fires. Recognizing the influential role of the internet in education, the research developed an e-learning system on disaster preparedness specifically designed for children. The system presented disaster-related content in an interactive, gamified format to make learning engaging while imparting essential safety knowledge for both natural and man-made emergencies. It featured information modules, topic-based assessments, and multimedia resources, such as videos and animations, to enhance understanding. The study also evaluated the effectiveness of these digital materials in improving children's disaster preparedness knowledge. (Manalang, 2020).

Synthesis

The reviewed literature, studies and theories emphasizes the transformative potential of gamification and interactive simulations in addressing persistent gaps in disaster preparedness education, particularly among secondary highschool learners in highly vulnerable countries such as the Philippines. Disaster risk reduction (DRR) initiatives in traditional school settings often rely on lectures and occasional drills, which, while informative, may fail to instill the quick decision-making skills and procedural memory necessary during real emergencies.

Studies by Bai (2024) and Arinta (2020) affirm that integrating game elements such as points, levels, challenges, and narratives into educational content significantly boosts learner motivation, engagement, and retention of critical knowledge. Gamified learning reframes disaster education from a passive, compliance-based activity into a dynamic, experience-driven process, where learners actively participate in problem-solving tasks within realistic contexts.

Frameworks like EscapeED (Clarke & Arnab, 2015) provide a structured foundation for designing educational escape rooms that align gameplay with learning objectives. By combining narrative immersion, time-bound challenges, and collaborative problem-solving, this approach not only reinforces theoretical understanding but also cultivates practical competencies. The Disaster Doors system builds on these principles, simulating earthquake, flood, and fire scenarios through an interactive 3D escape room format. This approach mirrors findings by Johnstun (2019) and Garguilo (2024), who observed that well-designed educational games promote active engagement, increase knowledge retention, and enhance learners' confidence in applying safety protocols under pressure.

Research on safety and risk-awareness games, such as Velazquez's (2022) Gamified Crisis Framework (GCF), underscores the value of context-specific, culturally relevant simulations. These games integrate societal conditions, learner characteristics, and localized hazards to produce training experiences that feel authentic and relevant. Studies like Saptaputra et al. (2024) and Marahatta et al. (2024) demonstrate that interactive drills and scenario-based exercises yield measurable improvements in participants' readiness, skill accuracy, and situational awareness. Such findings validate the Disaster Doors approach, which immerses learners in realistic, consequence-driven virtual environments where correct actions—such as

executing “duck, cover, and hold” during earthquakes or safely shutting off electrical systems during floods—are essential to progressing in the game.

Digital technologies in disaster education further expand the possibilities for immersive learning. Works by Ryubikin et al. (2023) show that digital simulations bridge the gap between theoretical instruction and actionable skill application by providing learners with repeated, safe practice opportunities. These findings are particularly relevant for Disaster Doors, which leverages 3D graphics, realistic physics, and scenario branching to replicate the unpredictability of real disasters while maintaining a safe environment for trial and error. The integration of ISO-based standards for earthquake safety, flood response, and fire management ensures that the simulated learning experience meets internationally recognized preparedness criteria.

In general, the literature, studies and theories strongly supports the viability of gamified simulation tools as effective complements to traditional disaster education. By combining established pedagogical frameworks with interactive, narrative-driven gameplay, Disaster Doors seeks to transform passive knowledge into active readiness. It directly addresses the limitations of conventional methods by engaging learners cognitively, emotionally, and behaviorally. More importantly, it positions junior high school students as empowered agents of their own safety, equipping them with the competence and confidence to act decisively in real emergencies. In doing so, the system contributes not only to individual preparedness but also to the broader goal of fostering resilient communities capable of withstanding and responding to disaster events in both urban and rural contexts.

Conceptual Framework

Figure 1 presents the conceptual framework, based on the Input-Process-Output. The input stage will involve data gathering through structured questionnaire, and learners' profile to understand the preferences of those who played the simulation. The Process stage development, based on V-model includes the shrunken version of the following stages: (1) Requirements Analysis; (2) System Design; (3) Architectural Design; (4) Module/Component Design; (5) Unit Testing; (6) Integration Testing; (7) System Testing, and; (8) Acceptance Testing. The output stage will be the development of Disaster Doors – it is a gamified disaster preparedness to add knowledge about disaster and prepared learners for the future events.

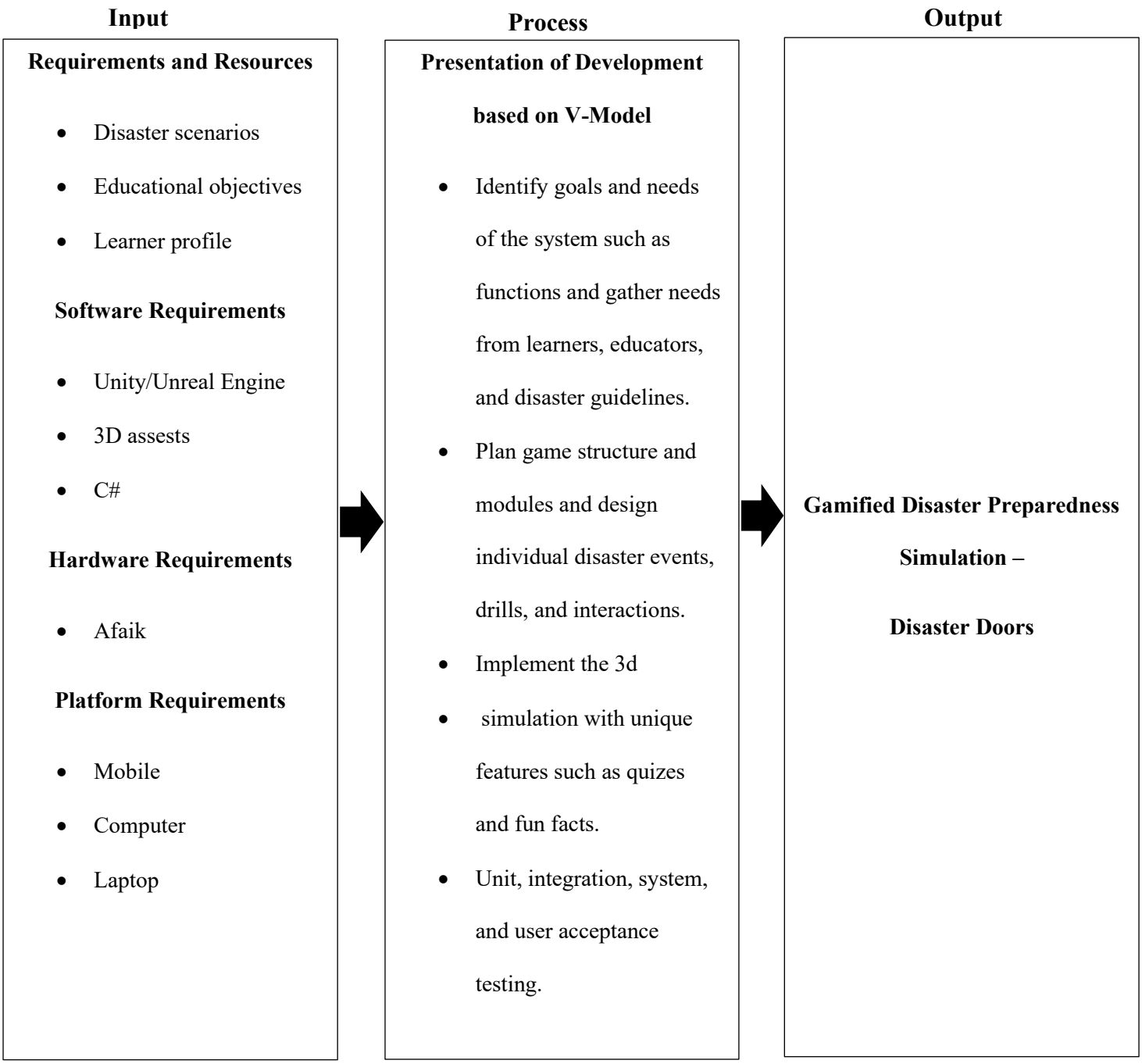


Figure 1. Conceptual Framework of the Study Using DisasterDoors Framework

The first frame shows the Input, which is the knowledge requirements to develop the systems' functionality. Moreover, it shows the required software, hardware, and platform for accessing the game.

The second frame presents the Process of shrunken development phases of V-model:

(1) Requiement Analysis; (2) System and Design; (3) Implementation, and; (4) Testing and Validation. With the support of the model, this game prioritizes clarity of requirements, structured development, early testing, and validated learning outcomes, making it reliable as an educational tool.

The third frame indicates Output, which is the result of the study – Disaster Doors.

The arrow lines connect the first frame (Input) to the second (Process), and to the third frame (Output) showing the logical flow of the study. This sequence illustrates how the initial elements of the study are transformed through systematic procedures to produce results that address the research objectives.

Definition of Terms

To provide clarity and avoid misinterpretation, the following terms are defined as they are specifically used in this study. These definitions are contextualized to show how each term relates to the development, implementation, and evaluation of Disaster Doors, the gamified disaster preparedness system created by the researchers.

Motivation – In this study, motivation refers to the willingness of learners to engage with the Disaster Doors game. It is important because a student's level of motivation

directly affects how actively they participate in disaster preparedness activities within the simulation.

Accessibility – This pertains to how easily learners can use and access the game. In the context of this research, accessibility means ensuring that the system can be run on devices available to students, such as computers or mobile phones, without causing barriers that may affect learning.

Applicability – Applicability refers to how the knowledge and skills gained from playing Disaster Doors can be applied in real-life disaster situations. For this study, it emphasizes whether the lessons from the game translate into practical responses during earthquakes, floods, or fires.

Disaster – In this study, disaster refers specifically to the three types of hazards simulated in the game: earthquake, flood, and fire. These disasters were chosen because they are among the most common and destructive hazards experienced in the Philippines.

Disaster Doors – Disaster Doors is the main system developed in this research. It is a gamified 3D learning simulation that uses "trial doors" to represent different disaster scenarios, aiming to enhance preparedness skills among junior high school students.

Educator – In the context of this study, educators are teachers who may integrate the game into their disaster risk reduction lessons to support student engagement and improve learning outcomes.

EscapeED Framework – This framework serves as the study's theoretical guide for designing the educational escape-room style gameplay in Disaster Doors. It helps align the game's mechanics with specific learning objectives related to disaster preparedness.

Game Delivery – Game delivery in this research refers to how instructions, flow, and activities are presented to learners during gameplay. A clear and effective delivery ensures that players understand how to complete tasks that reflect real-life disaster responses.

Game Design – This term refers to the visual and interactive features of Disaster Doors, such as graphics, audio, and animations, that make the game engaging for learners and encourage them to continue playing while learning preparedness skills.

Game Mechanics – In this study, game mechanics are the rules, objectives, and challenges integrated into Disaster Doors. These mechanics guide learners in solving problems and making decisions that mirror disaster response actions.

Learner – Learners in this study refer to junior high school students who are the target users of the Disaster Doors system. They are the primary participants in testing the effectiveness of the game in teaching disaster preparedness.

Realism – Realism describes how closely the disaster events inside the game resemble actual emergencies. In this study, realism is significant because realistic scenarios improve learners' ability to transfer what they learn in the game to real-world situations.

Trial Doors – Trial Doors are the in-game elements that represent earthquakes, floods, and fire challenges. Each door immerses the learner in a simulated environment where they must perform correct preparedness actions.

Usability – Usability refers to how easily students can navigate and interact with the game system. In this study, it is measured through learners' ability to understand controls, follow instructions, and successfully complete tasks within the simulation.

CHAPTER III

METHODS OF RESEARCH

This chapter outlines the research methods adopted in this study. It provides a comprehensive explanation of the procedures and strategies used to gather and analyse data relevant to the research objectives. Specifically, this chapter discusses the study of methods and techniques, population and sample, sampling design, research instrument, data gathering procedures, ethical considerations, and data analysis. Each component is carefully explained to ensure the reliability, validity, and ethical integrity of the study.

Methods and Techniques of the Study

The researchers employed a descriptive quantitative research design, which aims to collect and analyze numerical data to determine whether the Disaster Doors gamified escape room experience influences learners' resilience and preparedness for disasters. This approach was chosen to measure changes in learners' awareness, motivation, and readiness after participating in the activity.

Descriptive Method. The study utilized four instruments to gather data systematically. A structured questionnaire using a 5-point Likert scale was employed with two categories to ensure the essence of the study. Firstly, the first category was implemented to measure system's effectiveness in terms of game design, and mechanics . Lastly, the second part are questions on disaster preparedness concepts, was administered reflecting to their before and after simulation activity to determine knowledge acquisition, skill application and decision-making during simulated scenarios that will measure—what extent does the system improve disaster preparedness of learners. Using these methods allowed the researchers to obtain both

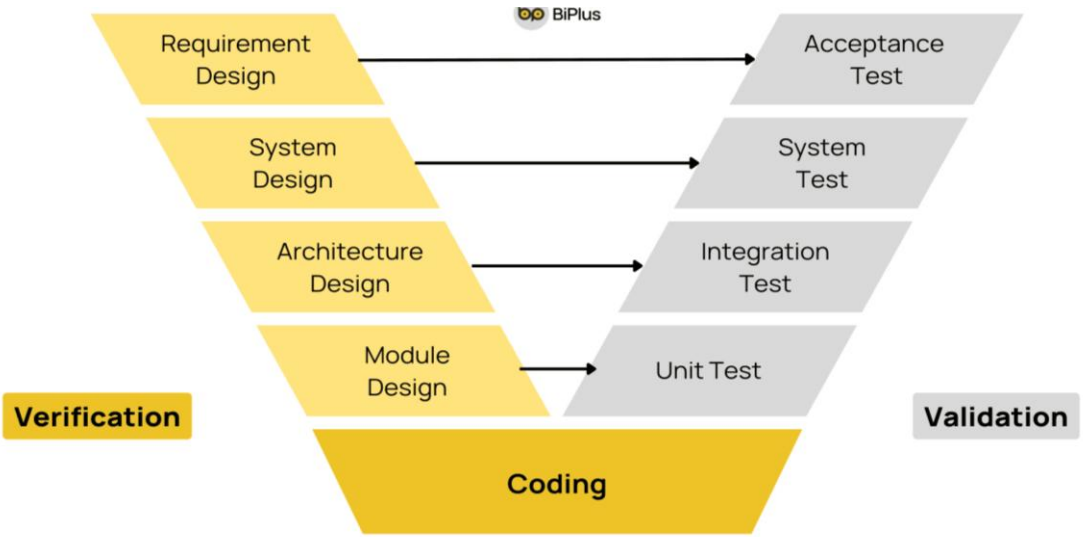
objective and subjective measures, producing a comprehensive evaluation of the game's effectiveness through the simulation in fostering disaster preparedness among learners.

Developmental Method. The developers used the system development process based on the V-model methodology for testing and planning to prioritize accuracy and validation of learning goals. Most importantly, designing the requirements of the system's three major disaster scenarios and defining learners' objectives by assessing decision-making during simulated situations. The acceptance testing then validated these by checking whether learners' pre- and post-assessments demonstrated increased knowledge acquisition and skill application in disaster preparedness. Moreover, the second stage's goal is to evaluate the functionality and overall architecture of the system. This stage ensures that the intended design works properly, particularly the time-limited challenges reflecting the disaster scenarios, while the testing phase evaluates whether the game elements perform as intended. The third stage is used for smooth flow between gameplay and research instruments; it focuses on dividing the system into modules—including game mechanics and system pre/post-assessment, and defining interactions among them. Additionally, individual modules or components are tested. For example, timers should start and stop correctly, correct answers should trigger positive feedback, and wrong actions should result in penalties. Furthermore, the developers used Unity/Unreal to code the 3D environment, and testing included running the system on different devices to check stability, responsiveness, and usability. Debugging was also performed to fix minor errors. This model was implemented to provide a structured and systematic development process that ensures each stage of the game's design is validated through corresponding testing, thereby guaranteeing the

reliability, accuracy, and educational effectiveness of Disaster Doors as a disaster preparedness tool.

Figure 2. V-Model Methodology

Figure 2 show the system's developmental method using V-Model. It is a development methodology shaped like the letter "V", where the left side represents the stages of development and design, the bottom represents implementation or



coding, and the right side represents the testing and validation. Each developmental phase has corresponding stages, on the left: Requirement Design, System Design, Architecture Design, and Module Design. Additionally, on the right side: Acceptance Test, System Test, Integration Test, and Unit Test.

Population and Sample of the Study

The target population of this study consists of secondary school learners, specifically junior high school students, from whom the sample was drawn. The selected participants were expected to provide the data necessary to address the research

objectives. A total of 100 junior secondary learners were randomly selected from various sections and grade levels of a private secondary school in Bulacan. These learners were chosen based on their potential to provide meaningful insights, as they were presumed to have sufficient knowledge to respond to the questions and relevant experience with interacting with digital tools, which are essential for engaging with the Disaster Doors gamified learning platform.

The sample size for this study was determined using Slovin's formula with a population of 900 learners and a margin of error of 5%. Using the formula:

$$n = \frac{N}{1 + Ne^2}$$

Wherein:

n is the sample size

N is the population size

e is a margin of error

The computed sample size is 277 learners. However, due to feasibility, the researchers used 100 respondents selected through convenience sampling.

Distribution of Respondents

Respondents	Total	Percent
7th grader	25	25%
8th grader	25	25%
9th grader	25	25%
10th grader	25	25%

Total	100	100%

Table 1. Distribution of Respondents

Sampling Design

The researchers use the convenience sampling technique for this study, which is used to select individuals who are available at the time and willing to participate – this includes individuals from different cluster such as section and year level. For the structure questionnaire, the researchers selected individuals who have experienced or took part of the study, participating in Disaster Doors. This ensures tha the participant can provide the needed information regarding their experiences, engagement, and knowledge gain related to the simulation. The respondents must follow the following criteria: Have completed the simulation, willing to participate, a secondary learner, and provided with consent.

Research Instrument

This section discusses the research instrument used by the researchers to collect key data for the study. These tools made collecting critical information needed for data preservation and decision-making from Junior High School students. The researchers used the following research instruments

To achieve this, the study utilized the following data collection techniques to have the final output and reliable solutions:

Survey: The researchers used the Likert scale collection tool to assess the performance of Disaster Doors regarding usability, accessibility, motivation, applicability, realism, and perceived preparedness. A five-point technique was applied to determine the respondents' level of agreement and disagreement. The point system was as follows: 5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, and 1 – Strongly Disagree. The researchers also used Google Forms to distribute and collect student survey responses.

Questionnaire: The researchers also used structured questionnaires to gather information from selected junior high school learners before and after their exposure to Disaster Doors. The questionnaire consisted of both open-ended and closed-ended questions. Open-ended questions allowed participants to share detailed experiences and feedback, while closed-ended questions provided measurable data on their disaster preparedness knowledge and the game's effectiveness. This instrument was designed to evaluate the learners' knowledge, skills, and decision-making abilities related to earthquake, flood, and fire response scenarios.

Knowledge Test: A pre-test and post-test were conducted to measure the learners' knowledge gain. The pre-test assessed the baseline understanding of disaster preparedness prior to playing the game, while the post-test measured improvements in knowledge and skill application after engaging with the simulation. This method enabled the researchers to determine the effectiveness of *Disaster Doors* in enhancing disaster preparedness education.

Data Gathering Procedures

The researchers prepared a letter of approval to conduct surveys and distribute questionnaires, signed by the capstone adviser. The pre-test questionnaires were administered to the selected junior high school students to measure their existing knowledge and preparedness. After the students played the Disaster Doors simulation, the post-test questionnaires and survey were distributed electronically through Google Forms. Participants were informed of the purpose of the study, the confidentiality of their responses, and the voluntary natureof their participation. The collected data were then consolidated and analyzed using descriptive statistics to evaluate the effectiveness of the system in terms of learners' motivation, usability, accessibility.

Ethical Considerations

Research Subject Informed Consent and Participation

The following is the process of informing consent and participation to the participants with respect to The Data Privacy Act of 2012 (Republic Act No. 10173) to protects the privacy of individuals' personal information while promoting innovation and growth:

In research, handling consent to inform participants is important. Particiapnts must read, aware and voluntarily agreed to be part of the study. They have the right to be fully informed about the consent form and ask questions and clarifications before deciding to participate in this study or not. They are also free to ask questions during your participation in this research. Furthermore, they should be informed about the study's purpose, procedures, possible risks and discomforts, benefits, termination of the study, and authority to ensure the decisions they made is credible. This process ensure that the information they shared will treat with respect for their rights and autonomy..

Anonymity and confidentiality

The anonymity and confidentiality of the respondents involved in the study are kept in utmost priority for its privacy. They have the right to not include or share their names for keeping their identity hidden without revealing the names in data gathering, analysis and the findings of the study.

Data protection

The data collected were stored in a strict access control, granting only access to authorized personnel such as the developer and the researcher. Moreover, it is also required to regulary check backed up data to protect against loss or corruption. By implementing these measures, it can enhance the security and privacy of the data gathered throughout its lifecycle, from gathering to analysis.

Data Analysis

The collected data from the respondents using structured questionnaire using 5-point likert scale were analyzed using descriptive statistics to address each research question. Descriptive statistics to summarize learner responses and performance. Additionally, Inferential statistics to test whether differences and relationships are statistically meaningful.

Descriptive Statistics. It is used to summarize the data before testing for impact. The following procedure would determine the weighted mean for summarizing the central tendency of likert response; The standard deviation for measuring how spread out are responses from the mean, and; Frequency and percentage for showing how often each point was chosen.

The following are the formula for each statistical methods:

Weighted Mean Formula:

$$WM = \frac{\sum(f \times w)}{N}$$

Wherein:

F is the frequency of each response

w is weight of each scale point (1-5)

N is the total number of respondents

Furthermore, the following 5-point Likert scale was utilize in interpreting the responses.

Scale	Range	Verbal Interpretation
5	4.21 to 5.00	Very Effective/Strongly Agree
4	3.41 to 4.20	Effective/Agree
3	2.61 to 3.40	Moderately Effective/Undecided
2	1.81 to 2.60	Slightly Effective/Disagree
1	1.00 to 1.80	Not Effective/Strongly Disagree

Table 2. 5-Point Likert Scale Responses

Standard Deviation Formula:

$$SD = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}}$$

Wherein:

xi is each response value

$\bar{x}$ is mean value

N is the number of responses

Interpretation:

Low SD means responses are consistent.

High SD means responses are varied.

Frequency and Percentage Formula: ✓

$$\% = \frac{f}{N} \times 100$$

Wherein:

f is frequency

N is total number of respondents

Intrepetation:

Shows how many respondents fall into each category.

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